RAMA RIZKY BELROUZY HABIR

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PROFILE

Electrical Engineering undergraduate at Universitas Brawijaya focused on embedded firmware, robotics controls, and hardware–software integration. Hands-on experience developing STM32 firmware, actuator communication, PCB designs in KiCad, and ESP32-based IoT systems. Interested in precision motion control, robotics, and automated instrumentation.

EDUCATION

Universitas Brawijaya — B.Eng. in Electrical Engineering
2024 – Expected 2028 | GPA: 3.26/4.00

EXPERIENCE

Embedded & IoT Systems Engineer — AI Center, Brawijaya University | *March 2025 – Present*

- Engineered and deployed IoT data logging nodes using **ESP32 (C/C++)**, streaming multi-sensor telemetry to central servers via **MQTT/HTTP** protocols for real-time analysis.
- Implemented edge computing solutions utilizing **Raspberry Pi** as dedicated on-site processing units and mini-workstations for research prototypes.
- Managed on-premise server virtualization using **Proxmox VE**, orchestrating containerized services **(Docker / LXC)** to host telemetry databases and AI deployment pipelines.
- Conducted live hardware-software demonstrations at technology exhibitions and the **Brawijaya Open House**, communicating embedded system architectures and AI capabilities to academic stakeholders and public visitors.

Hardware & Power Electronics Engineer — BRONE Mascot Robot, Brawijaya University | *August 2026 – Present*

- Architected and fabricated a custom multi-cell Li-ion/LiPo **Battery Management System (BMS) PCB** in KiCad, providing reliable high-current power delivery for the official mobile mascot robot.
- Engineered multi-stage hardware protection circuitry, integrating Over-Voltage (OVP), Under-Voltage (UVP), Over-Current (OCP), and thermal shutdown via NTC thermistors to ensure operational safety.
- Conducted benchtop electrical testing, load profiling, and thermal analysis to validate power integrity and battery health under active robotic operation.

Software Engineer — KRSRI Brawijaya Robotics Team | *August 2025 – Present*

- Architected and deployed bare-metal/HAL firmware in **C/C++** on **STM32F411CEU6** for an autonomous hexapod search-and-rescue robot.
- Implemented multi-DOF Dynamixel servo communication via half-duplex UART, integrating inverse kinematics (IK) and PID-tuned gait algorithms for adaptive multi-terrain locomotion.
- Engineered an onboard computer vision pipeline on NVIDIA Jetson Nano using TensorRT, enabling real-time victim detection and localization at 15 FPS.
- Developed a robust UART/Serial telemetry protocol between Jetson Nano and STM32, synchronizing high-level navigation decisions with real-time low-level motor actuation.
- Conducted sensor calibration (IMU, distance sensors) and Hardware-in-the-Loop (HIL) testing, collaborating with cross-functional mechanical and electrical teams to optimize power distribution and chassis stability.
- Spearheaded STEM community outreach by mentoring elementary students at **SD Unggulan Al-Ya'lu Malang**, teaching fundamental robotics and guiding the development of an **ESP32-based mobile transporter robot** capstone project.

Team Lead & Embedded Hardware Engineer — PKM-KI, Brawijaya University | *January 2026 – May 2026*

- Lead development of LA-Braille, an electromechanical refreshable Braille display, targeting approximately 80% lower fabrication cost than commercial piezoelectric alternatives.
- Design custom KiCad PCBs using TBD62783/TBD62083 driver ICs and 74HC238D decoders for multiplexed actuator control.
- Develop a mechanically latched, 3D-printed pin mechanism designed to eliminate steady-state holding power; integrate Raspberry Pi-based image processing with OpenCV and Tesseract OCR.

PROJECTS

Rehabilitation Exoskeleton — Embedded Firmware & ROS 2 Integration | *September 2026 – Present*

- Developed real-time embedded firmware on STM32H753ZI (ARM Cortex-M7) and STM32G0B1RE microcontrollers for a robotic exoskeleton intended to support stroke rehabilitation.

- Implemented an SPI driver for the BNO085 IMU using the CEVA Sensor Hub Transport Protocol (SHTP), enabling 200 Hz acquisition of quaternion and Euler orientation data.
- Integrated IMU and force-sensitive resistor (FSR) sensing for joint-motion tracking and force measurement to support assistive movement monitoring.
- Implemented CAN bus communication between the embedded control system and RobStride 03 actuators for motor interfacing and control.
- Integrated embedded sensor telemetry with ROS 2 for system monitoring and higher-level robotics processing.
- Collaborated with medical faculty researchers to translate rehabilitation requirements into embedded sensing and robotic-system functionality.

Open Duck Bipedal Robot — Embedded Firmware & Actuation | *September 2026 – Present*

- Integrated ROS 2-based robotics software on a Raspberry Pi 4-equipped bipedal robot, with RViz2 for visualizing robot state and perception outputs.
- Developed a distributed real-time perception pipeline utilizing an NMS-free YOLOv26 model on a local edge server, streaming low-latency detection telemetry back to the Raspberry Pi 4.
- Integrated perception results into the robot's ROS 2 workflow to support visual awareness and future autonomous behaviors.

5-Axis Robotic Arm | *August 2026*

- Developed an STM32F411-based robotic manipulator with a custom two-layer KiCad controller PCB.
- Implemented a half-duplex UART driver with tri-state bus control for Dynamixel Protocol 1.0 communication at 1 Mbps.
- Implemented forward and inverse kinematics in C/C++ and built a browser-based WebGL simulator for trajectory visualization and validation.

Project & simulator: ramahabir.github.io/6dof-simulator/

Modular IoT Monitoring — CV BumiAji Sejahtera | *March 2026 – August 2026*

- Built a network of three ESP32 sensor nodes and a central gateway for greenhouse monitoring.
- Developed FreeRTOS-based sensor acquisition, ESP-NOW communication, and MQTT telemetry; designed battery-charging and boost-regulation circuitry for off-grid operation.

Dashboard: devel-ai.ub.ac.id/agrinode/

Vacuum Frying Process Monitoring — CV BumiAji Sejahtera | *March 2026 – August 2026*

- Developed an edge-based monitoring system using Raspberry Pi 4 to monitor real-time temperature and pressure in industrial vacuum-frying equipment.
- Integrated thermocouple temperature sensing and pressure sensors to acquire process data and display live measurements through a local human–machine interface (HMI).
- Implemented batch-level data logging for potato and guava (*Psidium guajava*) frying processes to support production traceability and process monitoring.

Portable Soil Nutrient Monitoring Device — CV BumiAji Sejahtera | *March 2026 – August 2026*

- Integrated a portable multi-parameter soil sensor module with an Android smartphone via USB-C for field measurements and data logging.
- Supported soil analysis by collecting parameters including NPK, pH, electrical conductivity (EC), soil moisture, and temperature through a mobile interface.

HONORS & CERTIFICATIONS

- Gold Medal, International Technology Category — Online Asian Agrocomplex Student Competition (OAASC), 2025.
- Juara Harapan I — 2nd International Student Summit, International Technology Innovation Category, 2025.
- Juara Harapan II — Petroleum Research & Innovation to Magnify Engineers (PRIME), 2026.
- Microsoft Certified: Azure AI Fundamentals (AI-900) — Microsoft, 2025.

TECHNICAL SKILLS

Embedded Systems: C, C++, Python, STM32, ESP32, FreeRTOS, bare-metal firmware, peripheral drivers.

Robotics & Control: ROS 2, Gazebo Harmonic, RViz2, robot modeling, PID control, forward/inverse kinematics, Smart Servos (Dynamixel).s.

Electronics & Hardware: KiCad, PCB design, power regulation, UART, SPI, I2C, hardware debugging.

AI & Systems: Object Detection, NVIDIA Jetson Nano, Raspberry Pi, Git/GitHub, Docker, Proxmox VE.